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SOC CMM Assessment System

A web platform for running Security Operations Center maturity assessments — built on the SOC-CMM® framework, bilingual in English and Portuguese.

github.com/fernando-karl/soc_cmm_system

Built on the SOC-CMM® framework by Rob van Os · Not affiliated with soc-cmm.com

Measure how mature a SOC really is — and watch it improve

Guided assessments across the six SOC-CMM[®] domains, scored on the 0–5 maturity scale, with radar reporting per domain and repeat assessments that track progress over time.

Multi-client

Several organisations per account, each scoped to its owner.

Bilingual

The full interface in English and Brazilian Portuguese.

Self-hosted

FastAPI and SQLite. Docker, or one Python command.

The model was solid. The workflow was a spreadsheet.

BEFORE

- One workbook per client, emailed around
- No history between assessments
- Charts rebuilt by hand each time
- English only

WITH THE PLATFORM

- Guided, resumable questionnaire
- Every assessment kept and comparable
- Scoring and charts generated
- English and Portuguese

Six domains, thirty aspects, ninety-seven questions

Business

Strategy, governance, cost, privacy

People

Employment, training, knowledge

Process

Management, operations, use cases

Technology

SIEM, detection, analytics

Services

Catalog, hunting, vulnerabilities

Results

Success factors, reporting, sharing

Each answer maps to the SOC-CMM[®] maturity scale, **0 (Non-existent) through 5 (Optimizing)**, and rolls up per aspect and per domain.

THE WORKFLOW

A guided questionnaire, not a wall of questions

The assessment is broken into one domain at a time, then one aspect at a time. Progress is shown per aspect and per domain, answers save as you go, and an assessment can be left and resumed.

- Six numbered steps, one per domain
- Per-aspect completion percentages
- Guidance text from the framework on each question

The screenshot displays the 'SOC CMM Assessment' interface. At the top, a navigation bar includes 'SOC CMM', 'Home', 'Customers', 'Dashboard', 'Users', and a 'Save & Exit' button. Below the navigation bar, the assessment title 'SOC CMM Assessment' is shown, followed by 'Customer: Acme Financial | Organization: Acme Financial Group'. A progress bar indicates 'Step 1 of 6: Business Domain' and '1 of 9 questions answered (11%)'. A horizontal menu shows six steps: 1 Business, 2 People, 3 Process, 4 Technology, 5 Services, and 6 Results. The 'Business Domain' section is highlighted in blue. Below this, the instruction 'Select an aspect to evaluate:' is shown. Three aspects are listed: 'Business Strategy & Direction' (0 of 0 questions answered, 0%), 'Cost' (1 of 1 questions answered, 100%), and 'Charter' (0 of 1 questions answered, 0%).

SOC CMM Assessment

Customer: Acme Financial | Organization: Acme Financial Group

Step 1 of 6: Business Domain
1 of 9 questions answered (11%)

1 Business 2 People 3 Process 4 Technology 5 Services 6 Results

Business Domain
Business domain of SOC CMM

Select an aspect to evaluate:

Business Strategy & Direction
Business Strategy & Direction aspect of Business
0 of 0 questions answered 0%

Cost
Cost aspect of Business
1 of 1 questions answered 100%

Charter
Charter aspect of Business
0 of 1 questions answered 0%

Assessment: 2025 H2 Re-assessment | Completed: 2026-10-07 17:16:21

General Maturity of SOC

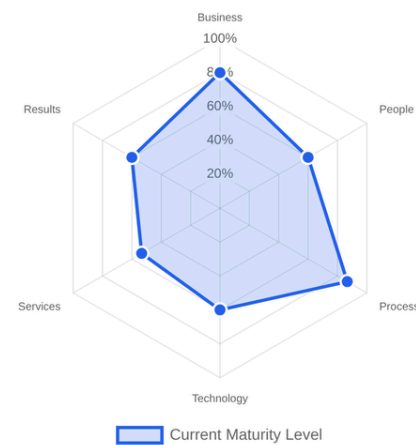


3.3/5.0

Average Maturity

Radar Chart of Maturity

Visualization of maturity across all domains



REPORTING

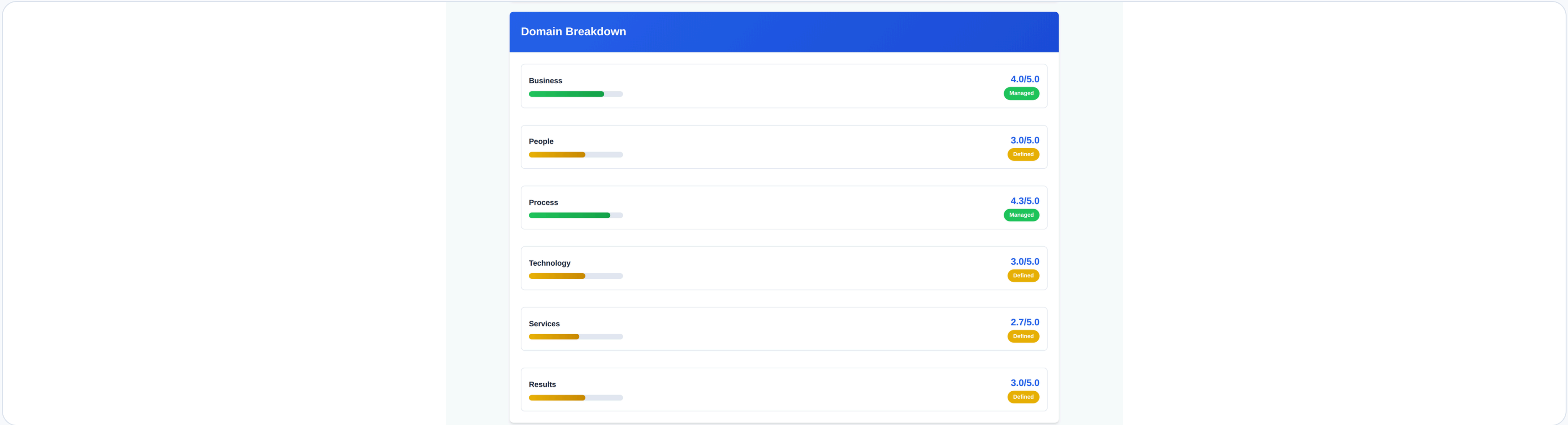
One number, then the shape behind it

An overall maturity score opens the report, and the radar chart immediately shows which of the six domains carry it and which drag it down.

Scores are computed from the maturity level of each answered option, averaged up through aspects to domains.

Every domain and aspect, with its maturity level named

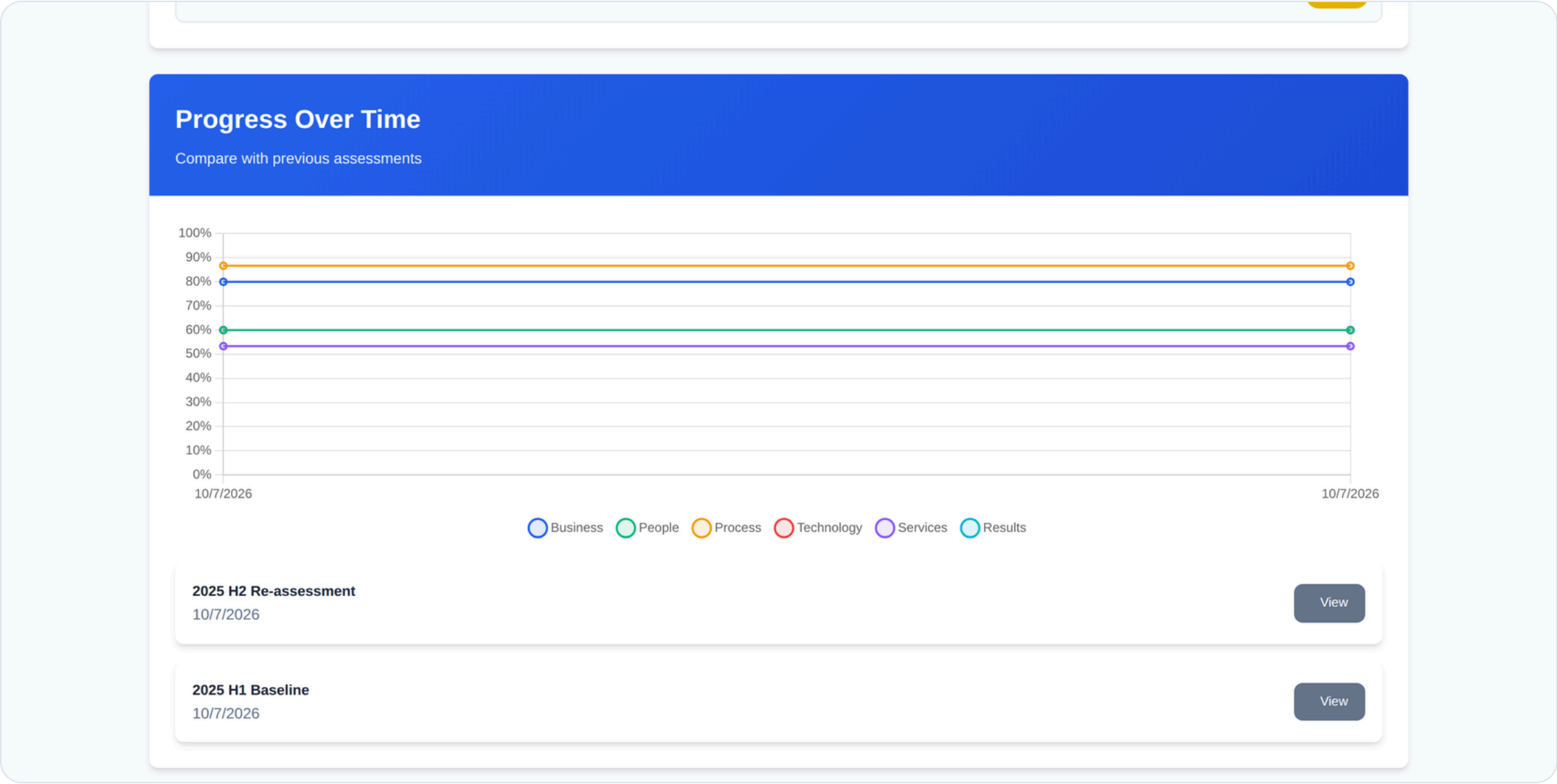
Scores are reported as a number and as the framework's own label — Defined, Managed, Optimizing — so a reader does not have to remember what 4.0 means.



OVER TIME

A maturity programme, not a one-off audit

Each completed assessment becomes a point per domain, so a re-assessment shows movement rather than a single new number.



ENGLISH · PORTUGUÊS

The language barrier is a real obstacle to adoption

SOC-CMM® is published in English. For many teams in Brazil that alone is enough friction to stop an assessment before it starts.

The entire interface — navigation, questionnaire, results and the attribution notice — switches between English and Brazilian Portuguese.

The screenshot displays the 'SOC CMM Assessment' web application. At the top, a navigation bar includes the 'SOC CMM' logo, links for 'Home', 'Clientes', 'Dashboard', and 'Usuários', and a language toggle set to 'Port' (Portuguese) next to 'Eng'. A 'Demo Analyst' button is also present. The main header shows 'SOC CMM Assessment' and a 'Salvar & Sair' button. Below this, the client information is 'Cliente: Acme Financial | Organização: Acme Financial Group'. A progress bar indicates 'Passo 1 de 6: Business Domínio' with '1 de 9 questões respondidas (11%)'. A step indicator shows six steps: 1 Business, 2 People, 3 Process, 4 Technology, 5 Services, and 6 Results. The 'Business Domínio' section is highlighted in blue. Under the heading 'Selecione um aspecto para avaliar:', three assessment areas are listed: 'Business Strategy & Direction' (0 de 0 questões respondidas, 0%), 'Cost' (1 de 1 questões respondidas, 100%), and 'Charter' (0 de 1 questões respondidas, 0%).

A REST API, and an MCP server for AI assistants

REST API

Clients, assessments, the questionnaire, answers and scores are all reachable over HTTP with a bearer token — enough to drive the whole workflow from a script.

MCP server

A Model Context Protocol server lets an AI assistant create a client, walk through the questionnaire, record answers and read back the scores.

```
export API_TOKEN='<bearer token>'  
python mcp_server.py
```

ACCESS CONTROL

A client's SOC weaknesses are sensitive data

Assessment results describe exactly where an organisation's defences are weakest. Every customer- and assessment-scoped endpoint requires authentication and verifies that the record belongs to the caller.

Scoped per owner

One ownership check, called from every endpoint that touches client data.

Proven by tests

A second tenant gets 403, an anonymous caller 401 — asserted on every endpoint.

Run in CI

The suite builds its own database and runs on every push.

BUILT ON SOC-CMM®

A derivative work, licensed the same way

The domains, aspects, questions and maturity levels come from the **SOC-CMM® framework v2.3.3 (basic)** by **Rob van Os**, published under CC BY-SA 4.0.

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Service IT Security supports organizations at every step, from conducting assessments to defining roadmaps and ensuring sustainable improvements in SOC operations.

Attribution & License

Credit for the SOC-CMM® framework this platform is built on

This platform is a **derivative work** of the [SOC-CMM®](#) (Security Operations Center Capability Maturity Model), version 2.3.3 (basic), created by **Rob van Os, MSc**. The domains, aspects, questions, maturity levels and guidance text all originate from that framework.

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For the original framework, please visit [soc-cmm.com](#).

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Clone it, run it, contribute

Run it

```
git clone .../soc_cmm_system
pip install -r requirements.txt
python main.py
```

Or **docker compose up**. Python 3.11+, SQLite, no external services.

Help wanted

- A first-run database bootstrap
- Broader test coverage
- More languages beyond EN and PT-BR
- Export to PDF and spreadsheet

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